

The Muller–Cohen–Matthews polynomial is a permutation polynomial

A human-readable account of the formalised proof

Abstract

We give a complete, human-readable proof of the following statement: for positive integers $k < n$ with $\gcd(k, n) = 1$ and k odd, the Muller–Cohen–Matthews (MCM) polynomial

$$M_{n,k}(z) = z \left(\sum_{i=0}^{k-1} z^{2^i-1} \right)^{2^k+1}$$

is a permutation polynomial of $L = \text{GF}(2^n)$. The proof follows Dobbertin’s route: one first proves the permutation criterion for the generalized Kasami polynomials q_α , and then transports it to the MCM side through the pair of mutually inverse linearized transfer maps $\psi_\beta, \varphi_\alpha$ (Dobbertin’s Theorem 4). Every step below corresponds to a machine-checked Lean statement of the accompanying development; the correspondence is tabulated in Appendix A.

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1 Setting and notation

Throughout, n and k are positive integers with

$$k < n, \quad \gcd(k, n) = 1,$$

and $L = \text{GF}(2^n)$ is the field with 2^n elements, of characteristic two. We write k' for a positive representative of the inverse of k modulo n , i.e.

$$k k' \equiv 1 \pmod{n}, \quad 0 < k' < n.$$

Since $\gcd(k, n) = 1$ such a k' exists, and together with it we fix the Bézout quotient $n' \in \mathbb{N}$ determined by

$$k'k = n'n + 1. \quad (1)$$

Two standing facts about L are used constantly and without further comment.

Lemma 1.1. *For all $x \in L$ we have $x^{2^n} = x$; consequently x^{2^r} depends only on $r \bmod n$. Moreover $u \mapsto u^{2^r}$ is a field automorphism, so $(u + v)^{2^r} = u^{2^r} + v^{2^r}$, and $-1 = 1$.*

Lemma 1.2. *For all $r \geq 1$, $\gcd(2^r - 1, 2^n - 1) = 2^{\gcd(r, n)} - 1$ and $\gcd(2^r, 2^n - 1) = 1$. In particular, since $\gcd(k, n) = 1$:*

1. $u \mapsto u^{2^k - 1}$ is a bijection of $L^\times$;
2. $u \mapsto u^{2^k}$ is a bijection of L , and $u^{2^k} = 1$ forces $u = 1$;
3. the kernel of the additive map $u \mapsto u^{2^k} + u$ is $\mathbb{F}_{2^{\gcd(k, n)}} = \{0, 1\}$.

The absolute trace polynomial is

$$\text{Tr}(z) = \sum_{i=0}^{n-1} z^{2^i},$$

an additive map $L \rightarrow \mathbb{F}_2 \subseteq L$ satisfying $\text{Tr}(z)^{2^r} = \text{Tr}(z)$ for all r .

Definition 1.3 (Generalized Kasami polynomial). For $\alpha \in \{0, 1\}$ put

$$q_\alpha(z) = \left(\sum_{i=1}^{k'} z^{2^{ik}} + \alpha \text{Tr}(z) \right) z^{(2^n - 1) - (2^k + 1)}.$$

Definition 1.4 (Generalized MCM polynomial). For $\beta \in \{0, 1\}$ put

$$P_\beta(z) = \left(\sum_{i=0}^{k-1} z^{2^i} + \beta \text{Tr}(z) \right)^{2^k + 1} z^{(2^n - 1) - 2^k}.$$

Definition 1.5 (MCM polynomial).

$$M_{n,k}(z) = z \left(\sum_{i=0}^{k-1} z^{2^i - 1} \right)^{2^k + 1}.$$

In each case we distinguish carefully between the *polynomial* and the *evaluation map* $L \rightarrow L$ it induces; “ p is a permutation polynomial” means that the evaluation map $x \mapsto p(x)$ is a bijection of L . As usual we use the convention $0^{-1} = 0$, so that $x \mapsto x^{-1}$ is a bijection of L fixing 0.

The goal of this note is:

Theorem 1.6 (Main theorem). *Let $0 < k < n$ with $\gcd(k, n) = 1$ and k odd. Then $M_{n,k}$ is a permutation polynomial of $\text{GF}(2^n)$.*

The proof occupies Sections 2–4: Section 2 proves the permutation criterion for q_α , Section 3 the transfer theorem relating q_α and P_β , and Section 4 deduces Theorem 1.6.

2 The permutation criterion for the generalized Kasami polynomials

The aim of this section is the following criterion, which is where all the work lies.

Theorem 2.1. *Let $0 < k < n$, $\gcd(k, n) = 1$, $0 < k' < n$ with $kk' \equiv 1 \pmod{n}$, and $\alpha \in \{0, 1\}$. Then q_α is a permutation polynomial of $\text{GF}(2^n)$ if and only if*

$$k' + \alpha n \equiv 1 \pmod{2}.$$

Throughout this section we abbreviate

$$S(z) = \sum_{i=1}^{k'} z^{2^{ik}}, \quad R(z) = S(z) + \alpha \text{Tr}(z),$$

so that $q_\alpha(z) = R(z)z^{(2^n-1)-(2^k+1)}$. Both S and R are additive maps $L \rightarrow L$.

2.1 Fibres of q_α as a Kasami equation

Lemma 2.2. *Let $c, z \in L$.*

1. *If $z \neq 0$ then $q_\alpha(z) = c$ if and only if*

$$c z^{2^k+1} = R(z). \tag{K}$$

2. *$q_\alpha(z) = 0$ if and only if $R(z) = 0$ (including for $z = 0$).*

Proof. Assume $z \neq 0$, so $z^{2^n-1} = 1$. Since $k < n$ we have $2^k + 1 \leq 2^n - 1$, hence multiplying $q_\alpha(z) = R(z)z^{(2^n-1)-(2^k+1)}$ by z^{2^k+1} gives $q_\alpha(z)z^{2^k+1} = R(z)z^{2^n-1} = R(z)$. As $z^{2^k+1} \neq 0$, $q_\alpha(z) = c$ is equivalent to (K). This proves (1), and (2) for $z \neq 0$ (take $c = 0$). For $z = 0$ both $q_\alpha(0)$ and $R(0)$ vanish, because every exponent occurring in S and Tr is positive. $\square$

2.2 Necessity of the parity condition

Proposition 2.3. *If q_α is a permutation polynomial then $k' + \alpha n \equiv 1 \pmod{2}$.*

Proof. Suppose $k' + \alpha n$ is even. Then

$$R(1) = \sum_{i=1}^{k'} 1 + \alpha \sum_{i=0}^{n-1} 1 = (k' + \alpha n) \cdot 1 = 0$$

in L , because L has characteristic two. By Lemma 2.2(2), $q_\alpha(1) = 0 = q_\alpha(0)$, while $1 \neq 0$ in L . Hence q_α is not injective. $\square$

2.3 Frobenius identities and the linearized equation

Lemma 2.4. *For every $x \in L$,*

$$S(x)^{2^k} = S(x) + x^{2^k} + x^{2^{k+1}}, \quad \text{Tr}(x)^{2^k} = \text{Tr}(x),$$

and consequently $R(x)^{2^k} = R(x) + x^{2^k} + x^{2^{k+1}}$.

Proof. Applying the automorphism $u \mapsto u^{2^k}$ termwise, $S(x)^{2^k} = \sum_{i=1}^{k'} x^{2^{(i+1)k}} = \sum_{i=2}^{k'+1} x^{2^{ik}}$. Hence, in characteristic two,

$$S(x)^{2^k} = S(x) + x^{2^k} + x^{2^{(k'+1)k}}.$$

By (1), $(k' + 1)k = k'k + k \equiv 1 + k \pmod{n}$, so by Lemma 1.1 $x^{2^{(k'+1)k}} = x^{2^{k+1}}$. The trace identity is the standard invariance of Tr under Frobenius, and the last claim follows since $\alpha \in \{0, 1\}$ satisfies $\alpha^{2^k} = \alpha$. $\square$

Definition 2.5. For $c \in L$ let

$$\ell_c(X) = c^{2^k} X^{2^{2k}} + X^{2^k} + cX + 1, \quad \ell_{0,c}(X) = \ell_c(X) + 1 = c^{2^k} X^{2^{2k}} + X^{2^k} + cX.$$

$\ell_{0,c}$ is $\mathbb{F}_2$ -linear, so $\ell_c(x) = \ell_c(y)$ if and only if $\ell_{0,c}(x + y) = 0$.

Proposition 2.6. *If $x \neq 0$ satisfies (K), then $\ell_c(x) = 0$.*

Proof. Raise (K) to the power 2^k and add (K) to it. Using Lemma 2.4 on the right-hand side,

$$c^{2^k} x^{2^{2k+2k}} + c x^{2^{k+1}} = R(x)^{2^k} + R(x) = x^{2^k} + x^{2^{k+1}}.$$

Dividing by $x^{2^k} \neq 0$ and using $x^{2^{k+1}}/x^{2^k} = x^{2^k}$ yields

$$c^{2^k} x^{2^{2k}} + c x + 1 + x^{2^k} = 0,$$

which is $\ell_c(x) = 0$. $\square$

Proposition 2.7. *Assume $k' + \alpha n \equiv 1 \pmod{2}$. If $R(x) = 0$ then $x = 0$.*

Proof. From $R(x) = 0$ and Lemma 2.4, $0 = R(x)^{2^k} + R(x) = x^{2^k} + x^{2^{k+1}} = x^{2^k}(1 + x^{2^k})$. Hence $x = 0$ or $x^{2^k} = 1$; in the latter case $x = 1$ by Lemma 1.2(2). But $R(1) = (k' + \alpha n) \cdot 1 = 1 \neq 0$ by the parity hypothesis. So $x = 0$. $\square$

2.4 First case: c is not of the form $\gamma^{2^k+1} + \gamma$

Lemma 2.8 (Factorisation). *Let $c \neq 0$ and $z \in L$. Put $D = cz^{2^k-1}$ and $A = D^{2^{n-1}}$, so that $A^2 = D$. Then*

$$\ell_{0,c}(z) = c^{-1}(A^{2^k+1} + A + c)^2 z.$$

Proof. Squaring is additive, so $(A^{2^k+1} + A + c)^2 = A^{2(2^k+1)} + A^2 + c^2 = D^{2^k+1} + D + c^2$. Now

$$c^{-1}zD^{2^k+1} = c^{-1}zc^{2^k+1}z^{(2^k-1)(2^k+1)} = c^{2^k}z^{2^{2k}}, \quad c^{-1}zD = c^{-1}zc^{2^k-1}z^{2^k-1} = z^{2^k},$$

and $c^{-1}zc^2 = cz$. Adding the three contributions gives exactly $\ell_{0,c}(z)$. (We used $A^2 = D^{2^n} = D$, which holds by Lemma 1.1.) $\square$

Proposition 2.9. *Suppose there is no $\gamma \in L$ with $c = \gamma^{2^k+1} + \gamma$. Then ℓ_c has at most one root in L .*

Proof. First, $c \neq 0$: otherwise $\gamma = 0$ would do. Let x, y be roots of ℓ_c and set $z = x + y$, so $\ell_{0,c}(z) = 0$ by linearity. If $z \neq 0$, Lemma 2.8 forces $A^{2^k+1} + A + c = 0$, i.e. $c = A^{2^k+1} + A$ (characteristic two), contradicting the hypothesis. Hence $z = 0$, i.e. $x = y$. $\square$

2.5 Second case: $c = \gamma^{2^k+1} + \gamma$

Assume now $c = \gamma^{2^k+1} + \gamma$ and $c \neq 0$; then $\gamma \neq 0$ and, since $c = \gamma(\gamma^{2^k} + 1)$, also $\gamma^{2^k} \neq 1$, i.e. $\gamma \neq 1$. Introduce

$$Q(X) = cX^{2^k} + \gamma^2 X + \gamma.$$

2.5.1 The auxiliary element Δ

Lemma 2.10. *There is a $\Delta \in L^\times$ with*

$$(\Delta^{2^k-1})^{-1} = \gamma^{2^k-1} + \gamma^{-1}, \quad (2)$$

and any such Δ satisfies

$$\Delta^{2^k} = \gamma\Delta + (\gamma\Delta)^{2^k}. \quad (3)$$

Proof. We have $\gamma^{2^k-1} + \gamma^{-1} = \gamma^{-1}(\gamma^{2^k} + 1) \neq 0$ because $\gamma \neq 1$. By Lemma 1.2(1) the map $u \mapsto u^{2^k-1}$ is a bijection of $L^\times$, so (2) has a solution $\Delta \in L^\times$. Rewriting (2) as $\Delta^{2^k-1}\gamma^{-1}(\gamma^{2^k} + 1) = 1$ and multiplying by $\gamma\Delta$ gives $\Delta^{2^k}(\gamma^{2^k} + 1) = \gamma\Delta$, which is (3). $\square$

Lemma 2.11. $\text{Tr}(\Delta) = 0$.

Proof. Apply Tr to (3). Since Tr is additive and Frobenius-invariant, $\text{Tr}(\Delta^{2^k}) = \text{Tr}(\Delta)$ and $\text{Tr}((\gamma\Delta)^{2^k}) = \text{Tr}(\gamma\Delta)$, so $\text{Tr}(\Delta) = \text{Tr}(\gamma\Delta) + \text{Tr}(\gamma\Delta) = 0$. $\square$

2.5.2 Factorisation of the zero set of ℓ_c through Q

Proposition 2.12. *For all $x \in L$,*

$$\ell_c(x) = Q(x)^{2^k} + (\Delta^{2^k-1})^{-1}Q(x),$$

and therefore

$$\ell_c(x) = 0 \iff Q(x) = 0 \text{ or } Q(x) = \Delta^{-1}. \quad (\text{F})$$

Proof. Write $\lambda = (\Delta^{2^k-1})^{-1} = \gamma^{2^k-1} + \gamma^{-1}$. Expanding,

$$Q(x)^{2^k} = c^{2^k}x^{2^{2k}} + \gamma^{2^{k+1}}x^{2^k} + \gamma^{2^k}, \quad \lambda Q(x) = \lambda c x^{2^k} + \lambda \gamma^2 x + \lambda \gamma.$$

Now $\lambda \gamma^2 = \gamma^{2^{k+1}} + \gamma = c$ and $\lambda \gamma = \gamma^{2^k} + 1$, whence the constant terms add up to $\gamma^{2^k} + \gamma^{2^k} + 1 = 1$ and the coefficient of x is c . Finally

$$\lambda c = (\gamma^{2^k-1} + \gamma^{-1})(\gamma^{2^{k+1}} + \gamma) = \gamma^{2^{k+1}} + \gamma^{2^k} + \gamma^{2^k} + 1 = \gamma^{2^{k+1}} + 1,$$

so the coefficient of x^{2^k} is $\gamma^{2^{k+1}} + \gamma^{2^{k+1}} + 1 = 1$. Altogether $Q(x)^{2^k} + \lambda Q(x) = c^{2^k}x^{2^{2k}} + x^{2^k} + cx + 1 = \ell_c(x)$.

For (F): $\ell_c(x) = 0$ means $Q(x)(Q(x)^{2^k-1} + \lambda) = 0$, i.e. $Q(x) = 0$ or $Q(x)^{2^k-1} = \lambda = (\Delta^{-1})^{2^k-1}$. By Lemma 1.2(1) the latter is equivalent to $Q(x) = \Delta^{-1}$. $\square$

2.5.3 Roots of Q do not solve the Kasami equation

Lemma 2.13. *Let x be a root of Q . Then*

1. $x^{2^k} = \gamma x + (\gamma x)^{2^k} + 1$, and more generally for all $i \geq 0$

$$x^{2^{(i+1)k}} = (\gamma x)^{2^{ik}} + (\gamma x)^{2^{(i+1)k}} + 1;$$

2. $\text{Tr}(x) = n \cdot 1$ (i.e. $\text{Tr}(x) = n \bmod 2$ in $\mathbb{F}_2 \subseteq L$);

3. $S(x) = \gamma x + (\gamma x)^2 + k' \cdot 1$;

4. $c x^{2^k+1} = \gamma x + (\gamma x)^2$.

Consequently

$$cx^{2^k+1} + S(x) + \alpha \operatorname{Tr}(x) = (k' + \alpha n) \cdot 1. \quad (4)$$

Proof. (1) $Q(x) = 0$ reads $(\gamma^{2^k+1} + \gamma)x^{2^k} + \gamma^2x + \gamma = 0$; dividing by $\gamma \neq 0$ gives $(\gamma^{2^k} + 1)x^{2^k} + \gamma x + 1 = 0$, i.e. $x^{2^k} = (\gamma x)^{2^k} + \gamma x + 1$. The general form follows by applying the automorphism $u \mapsto u^{2^{ik}}$ (note $1^{2^{ik}} = 1$).

(2) Apply $u \mapsto u^{2^j}$ to the case $i = 0$ of (1) and sum over $j = 0, \dots, n-1$. The left-hand side gives $\sum_j x^{2^{k+j}} = \operatorname{Tr}(x)$ by Lemma 1.1; the right-hand side gives $\operatorname{Tr}(\gamma x) + \operatorname{Tr}(\gamma x) + n \cdot 1 = n \cdot 1$.

(3) Summing the identity of (1) for $i = 0, \dots, k' - 1$, the middle terms telescope:

$$S(x) = \sum_{i=0}^{k'-1} x^{2^{(i+1)k}} = \gamma x + (\gamma x)^{2^{k'k}} + k' \cdot 1.$$

By (1), $k'k \equiv 1 \pmod{n}$, so $(\gamma x)^{2^{k'k}} = (\gamma x)^2$.

(4) Multiply $x^{2^k} = (\gamma x)^{2^k} + \gamma x + 1$ by x : $x^{2^k+1}(1 + \gamma^{2^k}) = \gamma x^2 + x$. Multiplying by γ and using $c = \gamma(\gamma^{2^k} + 1)$ gives $cx^{2^k+1} = \gamma^2x^2 + \gamma x = (\gamma x)^2 + \gamma x$.

Finally, adding (3), (4) and α times (2), the terms γx and $(\gamma x)^2$ cancel in characteristic two and (4) results. $\square$

Proposition 2.14. *Assume $k' + \alpha n \equiv 1 \pmod{2}$. Then no root of Q satisfies (K).*

Proof. If x satisfies (K) then $cx^{2^k+1} + S(x) + \alpha \operatorname{Tr}(x) = 0$, whereas if $Q(x) = 0$ then by (4) this same quantity equals $(k' + \alpha n) \cdot 1 = 1$. Since $1 \neq 0$ in L , the two are incompatible. $\square$

2.5.4 Uniqueness inside the fibre $Q(x) = \Delta^{-1}$

Lemma 2.15. *Let $x_0 \neq x_1$ satisfy $Q(x_0) = Q(x_1) = \Delta^{-1}$. Then*

$$x_0 + x_1 = \Delta, \quad \operatorname{Tr}(x_0) = \operatorname{Tr}(x_1), \quad S(x_0) + S(x_1) = \gamma\Delta + (\gamma\Delta)^2.$$

Proof. We first rewrite the fibre condition. Dividing $Q(x) = \Delta^{-1}$, i.e. $cx^{2^k} + \gamma^2x + \gamma = \Delta^{-1}$, by $\gamma^2\Delta$ and using

$$\frac{c}{\gamma^2\Delta} = \frac{\gamma^{2^k} + 1}{\gamma\Delta} = \frac{1}{\Delta^{2^k}}$$

(the last equality being (3) in the form $\Delta^{2^k}(\gamma^{2^k} + 1) = \gamma\Delta$), we find that $Q(x) = \Delta^{-1}$ is equivalent to

$$u + u^{2^k} = (\gamma\Delta)^{-1} + ((\gamma\Delta)^2)^{-1}, \quad u = x/\Delta.$$

Thus $u_0 = x_0/\Delta$ and $u_1 = x_1/\Delta$ have the same image under the additive map $u \mapsto u^{2^k} + u$, whose kernel is $\{0, 1\}$ by Lemma 1.2(3). So $u_0 = u_1$ or $u_0 = u_1 + 1$; as $x_0 \neq x_1$ the second holds, i.e. $x_0 + x_1 = \Delta$.

By Lemma 2.11 and additivity, $\operatorname{Tr}(x_0) + \operatorname{Tr}(x_1) = \operatorname{Tr}(\Delta) = 0$.

For the last identity, additivity of S gives $S(x_0) + S(x_1) = S(\Delta)$. Applying $u \mapsto u^{2^{ik}}$ to (3) yields $\Delta^{2^{(i+1)k}} = (\gamma\Delta)^{2^{ik}} + (\gamma\Delta)^{2^{(i+1)k}}$, and summing over $i = 0, \dots, k' - 1$ telescopes to

$$S(\Delta) = \gamma\Delta + (\gamma\Delta)^{2^{k'k}} = \gamma\Delta + (\gamma\Delta)^2,$$

using $k'k \equiv 1 \pmod{n}$ once more. (In contrast with Lemma 2.13(3) there is no constant term here, since (3) is homogeneous.) $\square$

For $x \in L$ set

$$e(x) = cx^{2^k+1} + S(x) + \alpha \operatorname{Tr}(x),$$

so that, by Lemma 2.2, x solves (K) exactly when $e(x) = 0$.

Proposition 2.16. *Let $x_0 \neq x_1$ satisfy $Q(x_0) = Q(x_1) = \Delta^{-1}$. Then $e(x_0) + e(x_1) = 1$.*

Proof. By Lemma 2.15 we may write $x_1 = x_0 + \Delta$, and

$$e(x_0) + e(x_1) = c(x_0^{2^k+1} + x_1^{2^k+1}) + \gamma\Delta + (\gamma\Delta)^2.$$

Expanding $x_1^{2^k+1} = (x_0 + \Delta)^{2^k}(x_0 + \Delta)$ in characteristic two,

$$x_0^{2^k+1} + x_1^{2^k+1} = \Delta x_0^{2^k} + \Delta^{2^k} x_0 + \Delta^{2^k+1} = \Delta x_0^{2^k} + \Delta^{2^k}(x_0 + \Delta).$$

From $Q(x_0) = \Delta^{-1}$ we get $c x_0^{2^k} = \Delta^{-1} + \gamma^2 x_0 + \gamma$, hence

$$c \Delta x_0^{2^k} = 1 + \gamma^2 \Delta x_0 + \gamma \Delta.$$

From (3), $\Delta^{2^k}(\gamma^{2^k} + 1) = \gamma\Delta$, and since $c = \gamma(\gamma^{2^k} + 1)$ this gives $c \Delta^{2^k} = \gamma \cdot \gamma\Delta = \gamma^2 \Delta$. Therefore

$$c(x_0^{2^k+1} + x_1^{2^k+1}) = 1 + \gamma^2 \Delta x_0 + \gamma\Delta + \gamma^2 \Delta(x_0 + \Delta) = 1 + \gamma\Delta + (\gamma\Delta)^2,$$

and adding $\gamma\Delta + (\gamma\Delta)^2$ leaves $e(x_0) + e(x_1) = 1$. $\square$

Proposition 2.17. *Assume $k' + \alpha n \equiv 1 \pmod{2}$ and $c = \gamma^{2^k+1} + \gamma$ for some $\gamma \in L$. Then (K) has at most one nonzero solution.*

Proof. If $c = 0$ then (K) reads $R(x) = 0$, and Proposition 2.7 gives $x = 0$; so there is no nonzero solution at all. Assume $c \neq 0$; then $\gamma \notin \{0, 1\}$ as observed above, and Δ is available. Let $x_0 \neq x_1$ be nonzero solutions of (K). By Proposition 2.6 both are roots of ℓ_c , so by (F) each satisfies $Q(x_j) = 0$ or $Q(x_j) = \Delta^{-1}$. Proposition 2.14 excludes the first alternative, so $Q(x_0) = Q(x_1) = \Delta^{-1}$. Proposition 2.16 then gives $e(x_0) + e(x_1) = 1$, contradicting $e(x_0) = e(x_1) = 0$. $\square$

2.6 Proof of Theorem 2.1

Proposition 2.18. *Assume $k' + \alpha n \equiv 1 \pmod{2}$. Let $c, x, y \in L$ satisfy (K) for x and for y , and suppose that $x \neq 0$ and $y \neq 0$ whenever $c \neq 0$. Then $x = y$.*

Proof. If $c = 0$, then $R(x) = R(y) = 0$ and Proposition 2.7 gives $x = y = 0$. Let $c \neq 0$, so $x, y \neq 0$. If c is not of the form $\gamma^{2^k+1} + \gamma$, then by Proposition 2.6 both x and y are roots of ℓ_c , and Proposition 2.9 gives $x = y$. If c is of that form, Proposition 2.17 gives $x = y$. $\square$

Proof of Theorem 2.1. Necessity is Proposition 2.3. For sufficiency assume $k' + \alpha n \equiv 1 \pmod{2}$ and let $q_\alpha(x) = q_\alpha(y) =: c$.

If $c = 0$ then by Lemma 2.2(2) $R(x) = R(y) = 0$, so $x = y = 0$ by Proposition 2.7. If $c \neq 0$ then $x \neq 0 \neq y$ (as $q_\alpha(0) = 0$), and by Lemma 2.2(1) both satisfy (K); Proposition 2.18 yields $x = y$. Hence the evaluation map of q_α is injective, and since L is finite it is bijective. Thus q_α is a permutation polynomial. $\square$

3 The transfer theorem

We now relate q_α and P_β through two linearized maps. Fix k', n' as in (1) and $\alpha, \beta \in \{0, 1\}$, and define $L \rightarrow L$ by

$$\psi_\beta(x) = \sum_{i=0}^{k-1} x^{2^i} + \beta \operatorname{Tr}(x), \quad \varphi_\alpha(x) = \sum_{i=0}^{k'-1} x^{2^{ik}} + \alpha \operatorname{Tr}(x).$$

Both are $\mathbb{F}_2$ -linear.

Lemma 3.1. For all $x \in L$, $R(x) = \varphi_\alpha(x)^{2^k}$.

Proof. $(\sum_{i=0}^{k'-1} x^{2^{ik}})^{2^k} = \sum_{i=0}^{k'-1} x^{2^{ik+1}} = S(x)$ and $\text{Tr}(x)^{2^k} = \text{Tr}(x)$, while $\alpha^{2^k} = \alpha$. $\square$

Lemma 3.2. For $x \neq 0$, $P_\beta(x) = \psi_\beta(x)^{2^k+1}/x^{2^k}$.

Proof. $P_\beta(x) = \psi_\beta(x)^{2^k+1}x^{(2^n-1)-2^k}$ by definition (note $2^k \leq 2^n - 1$ since $k < n$), and $x^{(2^n-1)-2^k} = x^{2^n-1}x^{-2^k} = x^{-2^k}$. $\square$

Proposition 3.3. Assume (1), $k' + \alpha n \equiv 1 \pmod{2}$ and $\beta \equiv n' + \alpha k \pmod{2}$. Then $\varphi_\alpha \circ \psi_\beta = \text{id}_L$; consequently ψ_β and φ_α are mutually inverse bijections of L .

Proof. Fix x and abbreviate $T = \text{Tr}(x)$. First,

$$\sum_{i=0}^{k'-1} \left(\sum_{j=0}^{k-1} x^{2^j} \right)^{2^{ik}} = \sum_{i=0}^{k'-1} \sum_{j=0}^{k-1} x^{2^{ik+j}} = \sum_{m=0}^{k'k-1} x^{2^m} = \sum_{m=0}^{n'n} x^{2^m} = x + n'T,$$

where the last step splits the range $\{0, \dots, n'n\}$ into the singleton $\{n'n\}$ contributing $x^{2^{n'n}} = x$ (Lemma 1.1) and n' blocks of n consecutive exponents, each contributing T . Second, $(\beta T)^{2^{ik}} = \beta T$, so the β -part of $\varphi_\alpha(\psi_\beta(x))$ contributes $k'\beta T$. Third,

$$\alpha \text{Tr}(\psi_\beta(x)) = \alpha \left(\sum_{j=0}^{n-1} \sum_{i=0}^{k-1} x^{2^{i+j}} + \beta \sum_{j=0}^{n-1} T \right) = \alpha(k + \beta n)T.$$

Altogether

$$\varphi_\alpha(\psi_\beta(x)) = x + (n' + \beta k' + \alpha k + \alpha \beta n)T,$$

and the integer coefficient is to be read modulo 2. Using $\beta \equiv n' + \alpha k$ and $k' + \alpha n \equiv 1$,

$$n' + \beta k' + \alpha k + \alpha \beta n = (n' + \alpha k) + \beta(k' + \alpha n) \equiv \beta + \beta \equiv 0 \pmod{2}.$$

Hence $\varphi_\alpha \circ \psi_\beta = \text{id}$. Since L is finite, both maps are bijections and each is the inverse of the other. $\square$

Theorem 3.4 (Transfer theorem; Dobbertin, Thm. 4). Assume $0 < k < n$, $\gcd(k, n) = 1$, $0 < k' < n$, (1), $k' + \alpha n \equiv 1 \pmod{2}$ and $\beta \equiv n' + \alpha k \pmod{2}$. Then

1. ψ_β and φ_α are mutually inverse bijections of L ;
2. $q_\alpha \circ \psi_\beta = (\cdot)^{-1} \circ P_\beta$ and $P_\beta \circ \varphi_\alpha = (\cdot)^{-1} \circ q_\alpha$;
3. q_α and P_β are both permutation polynomials of L .

Proof. (1) is Proposition 3.3.

(2) Let $x \in L$. If $x = 0$ then $\psi_\beta(0) = 0$, and $q_\alpha(0) = 0 = P_\beta(0)^{-1}$ (recall $0^{-1} = 0$), so the first identity holds. Let $x \neq 0$ and put $y = \psi_\beta(x)$; then $y \neq 0$, since $\varphi_\alpha(y) = x \neq 0 = \varphi_\alpha(0)$. By Lemma 2.2(1) applied to $c = q_\alpha(y)$, and by Lemma 3.1,

$$q_\alpha(y) y^{2^k+1} = R(y) = \varphi_\alpha(y)^{2^k} = x^{2^k}.$$

By Lemma 3.2, $P_\beta(x) = y^{2^k+1}/x^{2^k} \neq 0$, so $q_\alpha(y) P_\beta(x) = 1$, i.e. $q_\alpha(\psi_\beta(x)) = P_\beta(x)^{-1}$. The second identity follows by substituting $x \mapsto \varphi_\alpha(x)$ and using $\psi_\beta \circ \varphi_\alpha = \text{id}$.

(3) q_α is a permutation polynomial by Theorem 2.1, the parity hypothesis being satisfied. Hence $(\cdot)^{-1} \circ q_\alpha$ is a bijection of L , so by (2) $P_\beta \circ \varphi_\alpha$ is a bijection; since φ_α is a bijection, so is P_β . $\square$

4 The MCM polynomial

Lemma 4.1. *For $0 < k < n$, the polynomials $M_{n,k}$ and P_0 induce the same evaluation map on L .*

Proof. At $x = 0$ both vanish: $M_{n,k}(0) = 0$ because of the factor z , and $P_0(0) = 0$ because $\sum_{i=0}^{k-1} 0^{2^i} = 0$ (all exponents 2^i are positive) and $k \geq 1$.

Let $x \neq 0$, so $x^{2^n-1} = 1$. Since $2^i - 1 \geq 0$,

$$\sum_{i=0}^{k-1} x^{2^i} = x \sum_{i=0}^{k-1} x^{2^i-1} =: x \Sigma.$$

Therefore, using $2^k \leq 2^n - 1$,

$$P_0(x) = (x\Sigma)^{2^k+1} x^{(2^n-1)-2^k} = \Sigma^{2^k+1} x^{2^k+1+(2^n-1)-2^k} = \Sigma^{2^k+1} x x^{2^n-1} = x \Sigma^{2^k+1} = M_{n,k}(x). \quad \square$$

Proof of Theorem 1.6. Let $0 < k < n$, $\gcd(k, n) = 1$ and k odd. Choose k' with $0 < k' < n$ and $kk' \equiv 1 \pmod{n}$, and let n' be as in (1), i.e. $k'k = n'n + 1$. Apply Theorem 3.4 with

$$\alpha \equiv n' \pmod{2}, \quad \beta = 0.$$

Its two parity hypotheses hold:

- Reducing (1) modulo 2 and using that k is odd gives $k' \equiv k'k \equiv n'n + 1 \pmod{2}$; hence

$$k' + \alpha n \equiv (n'n + 1) + n'n \equiv 1 \pmod{2}.$$

- $\beta = 0$ and, again since k is odd, $n' + \alpha k \equiv n' + n'k = n'(1+k) \equiv 0 \pmod{2}$, so $\beta \equiv n' + \alpha k \pmod{2}$.

Theorem 3.4(3) therefore shows that P_0 is a permutation polynomial of L , and by Lemma 4.1 the evaluation map of $M_{n,k}$ coincides with that of P_0 . Hence $M_{n,k}$ is a permutation polynomial of $\text{GF}(2^n)$. $\square$

Remark 4.2. The hypothesis that k is odd is used exactly twice, namely in the two parity verifications above; it is what makes the choice $\beta = 0$ (the “classical” MCM polynomial, without trace term) compatible with the permutation parity $k' + \alpha n \equiv 1$ of the Kasami side.

A Correspondence with the formal development

Every statement of this note corresponds to a machine-checked declaration of the accompanying Lean development (namespace `GeneralizedKasami`). In the Lean files the field is `GF n`, the parameters α, β are elements of `Fin 2` named `a`, `b`, and k', n', Δ are `kInv`, `nPrime`, `Delta`.

Here	Lean declaration	File
Tr	<code>tracePolynomial</code>	<code>basic.lean</code>
q_α , its evaluation	<code>genKasamiPol, genKasamiEval</code>	<code>basic.lean</code>
R	<code>equationRHS</code>	<code>basic.lean</code>
$\ell_c, \ell_{0,c}$	<code>ell, ell_0</code>	<code>basic.lean</code>
$Q, c = \gamma^{2^k+1} + \gamma$	<code>gammaPolynomial, gammaCoefficient</code>	<code>GammaSteps.lean</code>
S	<code>iteratedKasamiSum</code>	<code>GammaSteps.lean</code>
Δ	<code>IsDelta</code>	<code>DeltaSteps.lean</code>
$e(x)$	<code>eTerm</code>	<code>PairSteps.lean</code>
$P_\beta, \psi_\beta, \varphi_\alpha$	<code>genMCMpol, psi_beta, phi_alpha</code>	<code>genMCM.lean</code>
$M_{n,k}$	<code>MCMpolynomial, MCMeval</code>	<code>MCMCorollary.lean</code>
Lemma 1.1	<code>gf_frobenius_pow_card, gf_frobenius_mod</code>	<code>basic.lean</code>
Lemma 2.2	<code>evaluation_eq_iff_equation_of_ne_zero,</code> <code>evaluation_eq_zero_iff_equation</code>	<code>basic.lean</code>
Prop. 2.3	<code>permutation_implies_parity</code>	<code>basic.lean</code>
Lemma 2.4	<code>kasami_sum_frobenius, trace_sum_frobenius</code>	<code>basic.lean</code>
Prop. 2.6	<code>kasamiEquation_implies_linearized</code>	<code>basic.lean</code>
Prop. 2.7	<code>equationRHS_eq_zero_unique</code>	<code>basic.lean</code>
Lemma 2.8	<code>ell_0_eval_factorization</code>	<code>basic.lean</code>
Prop. 2.9	<code>linearized_unique_of_not_exists_gamma</code>	<code>basic.lean</code>
Lemma 2.10	<code>exists_delta, delta_frobenius_identity</code>	<code>DeltaSteps.lean</code>
Lemma 2.11	<code>trace_sum_delta_eq_zero</code>	<code>DeltaSteps.lean</code>
Prop. 2.12	<code>ell_eval_eq_gammaPolynomial_frobenius,</code> <code>ell_eval_eq_zero_iff_gammaPolynomial</code>	<code>EllSteps.lean</code>
Lemma 2.13	<code>gammaPolynomial_eq_zero_iff(_iterate),</code> <code>trace_sum_of_gammaPolynomial_root,</code> <code>iteratedKasamiSum_of_gammaPolynomial_root,</code> <code>gammaPolynomial_root_final_identity</code>	<code>GammaSteps.lean</code>
Prop. 2.14	<code>kasamiEquation_not_gammaPolynomial_zero</code>	<code>genkasamipol.lean</code>
Lemma 2.15	<code>gammaPolynomial_eq_delta_inv_iff,</code> <code>frobenius_add_eq_frobenius_add_iff,</code> <code>pair_sum_eq_delta, pair_trace_eq,</code> <code>iteratedKasamiSum_delta</code>	<code>DeltaSteps.lean</code> <code>PairSteps.lean</code>
Prop. 2.16	<code>e0_plus_e1_equals_one</code>	<code>PairSteps.lean</code>
Prop. 2.17	<code>kasamiEquation_unique_of_exists_gamma</code>	<code>genkasamipol.lean</code>
Prop. 2.18	<code>kasamiEquation_unique</code>	<code>genkasamipol.lean</code>
Theorem 2.1	<code>parity_implies_permutation,</code> <code>generalizedKasami_isPermutation_iff</code>	<code>genkasamipol.lean</code>
Lemma 3.1	<code>equationRHS_eq_phi_pow</code>	<code>genMCM.lean</code>
Lemma 3.2	<code>genMCMeval_eq_div</code>	<code>genMCM.lean</code>
Prop. 3.3	<code>transfer_maps_inverse</code>	<code>genMCM.lean</code>
Theorem 3.4	<code>dobbertin_theorem4</code>	<code>genMCM.lean</code>
Lemma 4.1	<code>MCMeval_eq_genMCMeval_zero</code>	<code>MCMCorollary.lean</code>
Theorem 1.6	<code>MCMpolynomial_isPermutation</code>	<code>MCMCorollary.lean</code>

The Bézout data (1) is produced in Lean by `exists_positive_inverse_coefficients` (`MCMCorollary.lean`).
The formal statement of Theorem 1.6 is

```

theorem MCMpolynomial_isPermutation
  (n k : Nat) (hn : 0 < n) (hk : 0 < k) (hkn : k < n)
  (hcop : Nat.Coprime k n) (hkodd : Odd k) :

```

`IsPermutationPolynomial (MCMpolynomial n k)`

(the natural-number type is written `Nat` here for typesetting reasons), where `IsPermutationPolynomial p` unfolds to `Function.Bijective p.eval`.